

Adelic Harmonic Analysis, Dynamical Systems, and Spectral Operators: Reusability Assessment for Invariant-Set Quantum Generators

Research Question 1: Tate-Iwasawa Harmonic Analysis Beyond the Static Case

John Tate's 1950 doctoral thesis and the independent work of Kenkichi Iwasawa established a canonical reformulation of the analytic continuation and functional equation of Hecke L -functions¹. The core mathematical achievement of the Tate-Iwasawa theory is the application of Pontryagin self-duality and Poisson summation to the adèle ring $\mathbb{A}_{\mathbb{Q}}$ and the idele group $\mathbb{I}_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^{\times}$ ¹. By integrating Schwartz-Bruhat test functions over idele class quasi-characters, Tate demonstrated that the functional equation of the Riemann zeta function $\zeta(s)$ is a direct consequence of the global additive self-duality of $\mathbb{Q} \subset \mathbb{A}_{\mathbb{Q}}$ under a fixed character¹. In evaluating whether Tate's additive-multiplicative self-duality framework has been extended to time-evolving or iterated dynamical systems, four primary research results define the boundary between static Fourier analysis and dynamical operator theory:

1. **Tate, J. T. (1950):** *Fourier Analysis in Number Fields and Hecke's Zeta-Functions* (Ph.D. thesis, Princeton University; published in J. W. S. Cassels & A. Fröhlich, eds., *Algebraic Number Theory*, Academic Press, 1967, pp. 305–347)⁴.

- *Plain-Language Summary:* Constructs global zeta integrals

$\zeta(f, \chi) = \int_{\mathbb{I}_{\mathbb{Q}}} f(x) \chi(x) d^{\times} x$ on the adèle ring using Schwartz-Bruhat test functions $f \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$ and idele class characters χ ². Proves the functional equation of L -functions via Fourier inversion on locally compact abelian groups, leveraging the fact that the discrete subgroup $\mathbb{Q}$ is its own annihilator inside the self-dual group $\mathbb{A}_{\mathbb{Q}}$ ¹.

- *Classification: Static / Structural Result.* The Fourier-analytic integration takes place over static topological measure spaces without an explicit time parameter,

map iteration, or dynamic flow generator¹.

2. **Knauf, A. (2013):** *The Spectrum of an Adelic Markov Operator* (arXiv preprint arXiv:1305.6410)⁹.
 - *Plain-Language Summary:* Constructs a discrete-time Markov chain and its associated transition operator acting on the rank-two free module over the integer adeles $\hat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, driven by the linear action of $\mathrm{SL}(2, \mathbb{Z})$ ⁹. Demonstrates that the real spectrum of this adelic transition operator exhibits a spectral gap, whereas the non-real spectrum concentrates on a circle of radius $1/\sqrt{2}$ in the complex plane⁹.
 - *Classification: Genuinely Dynamical / Operator-Theoretic Result.* Defines an explicit iterated transfer operator on an adelic configuration space and analyzes its spectral decomposition⁹.
3. **Lindenstrauss, E. (2010):** *Adelic Dynamics and Arithmetic Quantum Unique Ergodicity* (Current Developments in Mathematics, 2009(1), 111–139)¹¹.
 - *Plain-Language Summary:* Develops measure rigidity and ergodic theory for algebraic subgroup actions on homogeneous spaces of the form $\mathrm{SL}(2, \mathbb{Q}) \backslash \mathrm{SL}(2, \mathbb{A}_{\mathbb{Q}})$ ¹¹. Formulates the Koopman operator $Uf(\zeta) = f(\beta(\zeta))$ associated with modular geodesic flows and proves measure-rigidity bounds that settle arithmetic quantum unique ergodicity for Maass forms¹¹.
 - *Classification: Genuinely Dynamical / Operator-Theoretic Result.* Uses Koopman operators and measure-preserving flows, though restricted to homogeneous algebraic quotient spaces rather than deterministic invariant-set substrates¹¹.
4. **Dragovich, B., Khrennikov, A. Yu., Kozyrev, S. V., & Volovich, I. V. (2009):** *On p -Adic Mathematical Physics* (p -Adic Numbers, Ultrametric Analysis and Applications, 1(1), 1–17)¹⁴.
 - *Plain-Language Summary:* Surveys quantum mechanics and field theory formulated over p -adic and adelic spaces¹⁴. Constructs p -adic Schrödinger-type differential operators using Vladimirov fractional derivatives D^α and reviews adelic wavefunctions formed as restricted tensor products of local states¹⁴.
 - *Classification: Operator-Theoretic / Partially Static Result.* Formulates kinetic differential operators over ultrametric spaces, but the spectral analysis focuses on static harmonic modes rather than dynamical flow generators¹⁴.

Work / Result	Configuration	Primary Mathematical	Dynamics	Classification
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	Space	Tool	Type	
Tate (1950) [cite: 4]	Adele Ring $\mathbb{A}_{\mathbb{Q}}$, Idele Group $\mathbb{I}_{\mathbb{Q}}$	Pontryagin Self-Duality, Poisson Summation	None (Static Fourier Analysis)	Static / Structural [cite: 1, 6]
Knauf (2013) [cite: 9]	Adelic Module $(\hat{\mathbb{Z}})^2$	$SL(2, \mathbb{Z})$ Representation , Markov Transfer Operator	Discrete Iterated Markov Dynamics	Genuinely Dynamical [cite: 9, 10]
Lindenstrauss (2010) [cite: 11]	Homogeneous Quotient $SL(2, \mathbb{Q}) \backslash SL$	Adelic Measure Rigidity, Koopman Operator $\mathcal{U}$	Continuous Geodesic/Auto morphic Flow	Genuinely Dynamical [cite: 11, 12]
Dragovich et al. (2009) [cite: 14]	Adelic Hilbert Space $L^2(\mathbb{A}_{\mathbb{Q}})$	Vladimirov Derivatives D^{α} , Tensor Product Kernels	Static Quantum State Generation	Operator-The oretic (Static) [cite: 14, 15]

The technical transition from Tate's static harmonic analysis to a genuinely dynamical framework requires replacing the fixed integration measure $d^{\times}x$ on $\mathbb{I}_{\mathbb{Q}}$ with an automorphism or endomorphism $T : X \rightarrow X$ acting on an adelic phase space X^2 . While homogeneous dynamics (such as the Einsiedler-Katok-Lindenstrauss program) exploits the algebraic rigidity of subgroup actions on $G(\mathbb{A}_{\mathbb{Q}})/G(\mathbb{Q})$, it does not utilize Tate's additive-multiplicative Fourier duality to construct linear Koopman generators for general deterministic flows¹. Knauf's construction represents a rare instance where adelic structure directly dictates the spectrum of a transfer operator; however, the mechanism is intrinsically stochastic (Markovian) rather than a continuous, self-adjoint deterministic generator⁹.

Research Question 2: Technical Details of Connes' Adele Class Space Program

Alain Connes' noncommutative geometry approach to the Riemann Hypothesis (RH)

reformulates the zeros of $\zeta(s)$ as an absorption spectrum on the adèle class space

$X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ ¹⁶. Understanding the precise technical architecture of this program requires evaluating whether it yields a directly diagonalized, self-adjoint operator or remains anchored to explicit trace formulas¹⁷.

1. **Connes, A. (1999): Trace formula in noncommutative geometry and the zeros of the Riemann zeta function** (Selecta Mathematica (N.S.), 5(1), 29–106)¹⁶.
 - *Plain-Language Summary:* Constructs the adèle class space $X_{\mathbb{Q}}$ as a noncommutative quotient space¹⁶. Defines an action of the idele class group $\mathbb{C}_{\mathbb{Q}} = \mathbb{I}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ via scaling transformations¹⁷. Proves an explicit global trace formula where the nontrivial zeros $\rho = \frac{1}{2} + i\gamma$ appear as missing spectral lines (an absorption spectrum) in the spectral distribution of the scaling flow, conditioned on cutoffs in phase space¹⁷.
 - *Classification: Trace-Formula / Structural Framework.* Does not produce a single, directly diagonalized self-adjoint Hamiltonian whose point eigenvalues are γ ¹⁷.
2. **Connes, A., & Consani, C. (2017): Geometry of the Scaling Site** (Selecta Mathematica (N.S.), 23(3), 1803–1850)²³.
 - *Plain-Language Summary:* Extends the adèle class space construction to tropical and semiring geometry, defining the "Scaling Site" as a Grothendieck topos equipped with a structure sheaf over the tropical semiring $\mathbb{R}_{\geq 0}^{\max, \times}$ ²³. The scaling action of $\mathbb{R}_{+}^{\times}$ is realized as an arithmetic flow on the object of points of the site over the absolute field $\mathbb{F}_1$ ²³.
 - *Classification: Algebro-Geometric / Noncommutative Structural Result.* Focuses on category-theoretic foundations and algebraic site structures rather than linear functional analysis on Hilbert spaces²³.
3. **Connes, A., & Consani, C. (2020): Weil positivity and trace formula, the archimedean place** (Selecta Mathematica (N.S.), 27(4), Paper No. 78 / arXiv:2006.13771)¹⁷.
 - *Plain-Language Summary:* Translates the semi-local trace formula into a concrete Hilbert space setting at the archimedean place $\mathbb{R}$ ¹⁷. Utilizes cutoff projections P (infrared cutoff at scale Λ) and $\hat{P} = FPF^{-1}$ (ultraviolet cutoff via Fourier

transform F) to define a compressed scaling operator on Sonin spaces¹⁷. Shows that the positivity of Weil's explicit quadratic form corresponds to the unitarity of the scaling flow compressed onto the orthogonal complement of the cutoff projections¹⁷.

- *Classification: **Compressed Spectral / Operator-Theoretic Result.*** Employs explicit projection operators and Hilbert space compressions, but serves to demonstrate Weil positivity rather than generating a discrete linear point spectrum¹⁷.

4. **Connes, A., Consani, C., & Moscovici, H. (2024):** *Zeta zeros and prolate wave operators* (Annals of Functional Analysis, 15(4), Article 87)¹⁸.

- *Plain-Language Summary:* Applies rank-one perturbations to a spectral triple built from the scaling operator and finite Euler products¹⁸. Demonstrates that the lowest-lying eigenvalues of these self-adjoint operators agree numerically to high precision with the nontrivial zeros of $\zeta(s)$ ¹⁸.
- *Classification: **Genuinely Operator-Theoretic (Local/Approximative) Result.*** Reaches a concrete self-adjoint operator formulation with quantized point spectrum, though restricted to finite truncation limits¹⁸.

Architectural Component	Mathematical Realization	Primary Role in Program	Operator Type / Status
Adele Class Space $X_{\mathbb{Q}}$ [cite: 17]	$\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ Noncommutative Space	Phase space housing the scaling action	Quotient Space (Domain) ¹⁷
Idele Class Flow $\mathbb{C}_{\mathbb{Q}}$ [cite: 17, 20]	Multiplicative action of $\mathbb{I}_{\mathbb{Q}}/\mathbb{Q}^{\times} \cong \mathbb{R}_{+}^{\times} \times$	Generates the continuous scaling flow $f(x) \mapsto$	Unbounded Continuous Flow ¹⁷
Sonin Space Cutoffs [cite: 17, 24]	Projections $P_{\leq \Lambda}$ and $\hat{P}_{< \Lambda}$ on $L^2(\mathbb{R})$	Phase-space regularization suppressing continuous continuum	Bounded Self-Adjoint Projections ¹⁷
Prolate Wave Perturbations	Rank-one self-adjoint	Quantizes zeros into discrete point	Self-Adjoint Operator

[cite: 18]	perturbations of spectral triples	spectrum	(Discrete) ¹⁸
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The idele class group $\mathbb{I}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ plays a fundamental structural role that prevents trivial generalization beyond number fields⁵. Its decomposition into the archimedean ray $\mathbb{R}_{+}^{\times}$ and the compact group of norm-one idele classes $K = \mathbb{I}_{\mathbb{Q}}^1/\mathbb{Q}^{\times}$ ensures that the non-archimedean places (primes p) are tied to the continuous scaling parameter $\lambda \in \mathbb{R}_{+}^{\times}$ through the global product formula $\prod_v |x|_v = 1$ ⁵. Without this tight arithmetic coupling, an arbitrary dynamical system on an ultrametric space lacks the global boundary condition necessary to compress the continuum into a trace formula matching arithmetic primes¹⁷.

Research Question 3: Berry-Keating Operators and p-Adic Quantum Chaos

The Berry-Keating conjecture posits that the nontrivial zeros of $\zeta(s)$ correspond to the energy spectrum of a quantum Hamiltonian originating from the classical scaling operator $H = xp$ ²⁸. While originally formulated on continuous real phase space $\mathbb{R}^2$, extensive work has sought p -adic and adelic reformulations, alongside developments in p -adic quantum chaos and random matrix theory²⁹.

1. **Srednicki, M. (2011):** *The Berry-Keating Hamiltonian and the Local Riemann Hypothesis* (Journal of Physics A: Mathematical and Theoretical, 44(30), 305202 / arXiv:1104.1850)²⁹.

- Plain-Language Summary: Formulates the local Riemann hypothesis for both $\mathbb{R}$ and $\mathbb{Q}_p$ ²⁹. For the real case, proves that the imaginary parts of the zeros of the Mellin transform of harmonic-oscillator wavefunctions are precisely the eigenvalues of the Berry-Keating operator $\hat{H}_{BK} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ projected onto subspaces of lower-level oscillator states²⁹. Analyzes the p -adic analogue using p -adic harmonic oscillator state vectors, showing that local Mellin zeros satisfy $\text{Re}(s) = 1/2$ ²⁹.
- Classification: **Operator-Theoretic / Local Spectral Model.** Constructs a local p -adic version of the xp operator projected onto discrete oscillator subspaces²⁹.

2. **Sierra, G., & Townsend, P. K. (2008) / Sierra, G. et al. (2014):** *Landau levels, Rindler*

space-time, and Riemann zeros (Physical Review Letters, 101(11), 110201) / *The xp Hamiltonian revisited* (Journal of Physics A: Mathematical and Theoretical, 47(32), 325204)²².

- *Plain-Language Summary:* Reinterprets the Berry-Keating operator $\frac{1}{2}(xp + px)$ as the Hamiltonian of a massless Dirac fermion in 1+1 dimensional Rindler space-time, bounded by moving mirrors situated at prime-logarithm positions²². The phase shifts induced by boundary conditions match the smooth phase of the Riemann zeta function²².
- *Classification:* **Genuinely Dynamical / Relativistic Field Model.** Provides a spatial-boundary realization of xp , though reliant on real space-time coordinates rather than an underlying adelic space³⁰.

3. **Van Peski, R. (2026):** *Reflecting Poisson walks and dynamical universality in p -adic random matrix theory* (The Annals of Probability, 54(5), 2568–2625 / arXiv:2312.11702)³⁴.

- *Plain-Language Summary:* Establishes p -adic random matrix theory by analyzing products and corners of random matrices over $\mathbb{Q}_p$ under $\mathrm{GL}_n(\mathbb{Z}_p)$ invariance³². Proves that the singular numbers of p -adic matrix products converge dynamically to a universal interacting particle system: the *reflecting Poisson sea*³². This serves as the exact p -adic analogue of Dyson Brownian motion, Airy/sine line ensembles, and quantum ergodicity in random matrix theory³².
- *Classification:* **Genuinely Dynamical / Stochastic Operator Theory.** Rigorously constructs p -adic spectral ensembles and dynamical flows, providing technical tools for p -adic quantum chaos³².

4. **Anashin, V., Khrennikov, A., & Yurova, E. (2014):** *p -Adic Ergodic Theory and Measure-Preserving Mappings* (Discrete Mathematics and Applications, 24(2), 125–142)³⁹.

- *Plain-Language Summary:* Formulates criteria for p -adic ergodicity and measure preservation for 1-Lipschitz (compatible) functions $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ via transitivity modulo p^k ³⁹. Applies van der Put series expansions to construct measure-preserving dynamical systems on p -adic integer balls³⁹.
- *Classification:* **Genuinely Dynamical / Measure-Theoretic Result.** Focuses on deterministic discrete-time dynamical systems over ultrametric spaces³⁹.

Model / Construction	Configuration Space	Primary Mathematical Operator	Spectral Target / Outcome	Physical/Dynamical Context
Berry-Keating (1999) [cite: 28, 29]	Real Phase Space $\mathbb{R}^2$	$\hat{H} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$	Smooth average of Riemann zeros ²⁸	Semiclassical Quantum Chaos ²⁸
Srednicki Local p-Adic (2011) [cite: 29]	p -Adic Field $\mathbb{Q}_p$	Projected x^p on p -adic Oscillator States	Zeros of p -adic Mellin transforms ²⁹	Local p -Adic Quantum Mechanics ²⁹
Sierra-Townsend (2008/2014) [cite: 30, 33]	1+1 Rindler Space-Time	Relativistic Dirac Hamiltonian with Mirrors	Bound-state match to zeta phases ²²	Accelerated Observer / Mirror Field Theory ³⁰
Van Peski p-Adic RMT (2026) [cite: 35]	Matrix Space $GL_n(\mathbb{Q}_p)$	Reflecting Poisson Sea Generator	p -Adic analogues of sine/Airy kernels ³²	p -Adic Dyson Brownian Motion ³²
Anashin-Khrennikov (2014) [cite: 39]	p -Adic Integers $\mathbb{Z}_p$	Perron-Frobenius / Koopman shift $f(x)$	Ergodic uniform measure preservation ³⁹	Deterministic Ultrametric Dynamics ³⁹

Research Question 4: Singular Continuous Spectrum in the Literature

In quantum mechanics and functional analysis, the spectrum $\sigma(H)$ of a self-adjoint operator decomposes into three mutually singular components: pure point spectrum (σ_{pp}), absolutely continuous spectrum (σ_{ac}), and singular continuous spectrum (σ_{sc})⁴¹. Singular continuous spectrum is characterized by continuous spectral measures that are supported on sets of

Lebesgue measure zero, such as Cantor sets or self-similar fractals⁴¹.

In the explicit operator constructions surrounding the Riemann Hypothesis, singular continuous spectrum is **entirely absent by design**¹⁷. The mathematical structure of RH requires that if a

Hilbert-Pólya operator H exists, its spectrum must be a **pure point spectrum** ($\sigma(H) = \sigma_{pp}(H) \subset \mathbb{R}$) corresponding precisely to the discrete imaginary parts γ_n of

the nontrivial zeros $\rho_n = 1/2 + i\gamma_n$ ¹⁸. In Connes' trace-formula framework, the background continuum is absolutely continuous, out of which the zeros are extracted as sharp absorption lines¹⁷. A singular continuous spectral component would indicate irregular,

non-quantized spectral fluctuations, which contradicts the discrete counting function $N(T)$ of the zeta zeros¹⁷.

In stark contrast to the RH literature, singular continuous spectrum is **widely prevalent and**

natural in mathematical physics when operators act on self-similar, fractal, or $\mathbb{P}$ -adic/ultrametric substrates⁴¹.

1. **Peherstorfer, F., Yuditskii, P., & Volberg, A. (2006):** *Limit periodic Jacobi matrices with a prescribed p -adic hull and a singular continuous spectrum* (Mathematical Research Letters, 13(2), 215–230)⁴².
 - *Plain-Language Summary:* Constructs infinite Jacobi matrices whose potential sequences are limit-periodic and whose topological hull is isomorphic to the ring of $\mathbb{P}$ -adic integers $\mathbb{Z}_{\mathbb{P}}$ ⁴². Proves that these operators exhibit purely singular continuous spectrum supported on zero-Lebesgue measure Cantor sets generated by polynomial iterations⁴².
 - *Classification: Operator-Theoretic / Spectral Result.* Establishes that $\mathbb{P}$ -adic topological hulls natively induce singular continuous spectra for tridiagonal discrete operators⁴⁴.
2. **Breuer, J. (2006):** *Singular Continuous Spectrum for the Laplacian on Certain Sparse Trees* (Communications in Mathematical Physics, 269(3), 851–857)⁴³.
 - *Plain-Language Summary:* Analyzes graph Laplacians Δ acting on rooted, spherically homogeneous metric and discrete trees (which are topologically equivalent to ultrametric $\mathbb{P}$ -adic spaces)⁴³. Proves that sparse tree structures systematically force the Laplacian to exhibit purely singular continuous spectrum, with explicit fractional Hausdorff dimensions assigned to the spectral measures⁴³.
 - *Classification: Operator-Theoretic / Graph Spectral Result.* Demonstrates that hierarchical, tree-like configuration spaces naturally suppress absolutely continuous scattering and point states in favor of singular continuous measures⁴³.
3. **Fan, A. H., Li, M. T., Yao, J. Y., & Zhou, D. (2020):** *Spectral measures in the field of*

p -adic numbers (arXiv preprint arXiv:2002.07559)⁴¹.

- *Plain-Language Summary*: Characterizes all spectral probability measures μ on $\mathbb{Q}_p$ that admit an orthonormal basis of continuous characters in $L^2(\mu)$ ⁴¹. Proves that singular spectral measures on $\mathbb{Q}_p$ are generically supported on p -homogeneous Cantor-like sets of Haar measure zero⁴¹.
 - *Classification: Harmonic Analysis / Measure-Theoretic Result*. Proves that p -adic harmonic analysis natively supports singular continuous spectral pairs⁴¹.
4. **Simon, B. (1996)**: *Operators with singular continuous spectrum: V. Sparse potentials / VI. Graph Laplacians* (Communications in Mathematical Physics, 181(3), 771–787)⁴⁷.
- *Plain-Language Summary*: Establishes that singular continuous spectrum, far from being a pathological anomaly, is generic (in the Baire category sense) for generic limit-periodic, almost-periodic, and sparse tree operators⁴².
 - *Classification: Operator-Theoretic Result*. Confirms that self-similar/fractal spatial substrates naturally select for singular continuous spectral types⁴².

Context / Domain	Primary Target Spectrum	Structural Mechanism	Presence of Singular Continuous Spectrum
Riemann Hypothesis / Hilbert-Pólya [cite: 18, 29]	Pure Point Spectrum (σ_{pp})	Discrete zero count $N(T)$, trace formula absorption ¹⁷	Explicitly Excluded / Absent [cite: 17, 18]
p-Adic Limit-Periodic Hulls [cite: 44]	Singular Continuous Spectrum (σ_{sc})	Cantor equilibrium measures on $\mathbb{Z}_p$ [cite: 44]	Generic / Native Feature [cite: 44]
Sparse Tree Graph Laplacians [cite: 43, 47]	Singular Continuous Spectrum (σ_{sc})	Hierarchical ultrametric branching ⁴³	Generic / Native Feature [cite: 43, 47]
p-Adic Spectral	Singular Continuous	p -Homogeneous	Rigorously Proven [cite: 41]

Measures [cite: 41]	Spectrum (σ_{sc})	Cantor sets in $\mathbb{Q}_p$ [cite: 41]	
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The mathematical physics literature demonstrates that seeking a closed, linear operator with a singular continuous spectrum on a p -adic or adelic invariant set is highly well-precedented outside of RH⁴¹. If a physics program targets a fractal invariant set, singular continuous spectrum represents the primary mathematical regime⁴¹.

Research Question 5: Direct Search for Existing Syntheses and Negative Results

To evaluate whether direct syntheses exist across these discipline boundaries, explicit pairwise and triple intersections were investigated across peer-reviewed mathematics and mathematical-physics literature.

Concept Intersection Matrix	Peer-Reviewed Status	Primary Literature / Findings
"Adelic Dynamics" ✕ "Koopman Operator"	Partial Match (Homogeneous Only)	Restricted entirely to arithmetic quantum unique ergodicity on homogeneous quotient spaces $SL(2, \mathbb{A}_{\mathbb{Q}})/SL(2, \mathbb{Q})$ (Lindenstrauss) ¹¹ . Zero literature applies this to deterministic Koopman operators on invariant sets.
"Tate's Thesis" ✕ "Koopman Operator"	NEGATIVE RESULT	No peer-reviewed papers exist. Tate's self-duality has never been combined with Koopman operator spectral theory ¹ .
"Berry-Keating" ✕ "p-Adic / Adelic Space"	Match (Local Only)	Srednicki (2011) connects local p -adic harmonic oscillators to Mellin transform zeros ²⁹ . No

		global adelic formulation of xp exists.
"Koopman Operator" × "Singular Continuous Spectrum" × "Adelic Space"	NEGATIVE RESULT	No peer-reviewed papers exist. The intersection of Koopman spectral theory, singular continuous measures, and adelic phase space is completely unstudied.
"Bohmian Mechanics" × "Adelic Invariant Set"	NEGATIVE RESULT	No peer-reviewed literature exists. No rigorous bridge connects Bohmian quantum-amplitude dynamics to an adelic invariant-set substrate.

During the survey of these thematic intersections, several online claims and preprints circulating on forums (e.g., Reddit r/LLMPhysics, viXra 2604.0024, Zenodo 18072982) were identified that claim to prove RH or unite these fields²⁶. These non-peer-reviewed claims assert the construction of "unitary adelic transfer operators" or "Haar measure self-duality proofs"

that reportedly force all non-trivial zeta zeros onto the critical line $\text{Re}(s) = 1/2$ ²⁶.

Technical evaluation reveals that these claims lack peer review and exhibit fundamental mathematical fallacies⁵⁰. Specifically, they conflate multiplicative Haar measure invariance on $\mathbb{I}_{\mathbb{Q}}$ with linear unitarity on $L^2(\mathbb{A}_{\mathbb{Q}})$, ignoring the non-unitary compression required by phase-space cutoffs¹⁷. These claims are formally classified as **false positives** and excluded from valid mathematical machinery⁵⁰.

Research Question 6: Reusable Machinery Independent of RH and Final Verdict

Independent of the truth or falsity of the Riemann Hypothesis, specific techniques from the surveyed literature can be evaluated for their potential reusability in constructing a linear spectral generator from a deterministic dynamical system:

1. Dual Phase-Space Cutoffs and Sonin Space Compression (Connes-Consani)¹⁷:

- *Technique:* Using dual phase-space projections $P_{\leq \Lambda}$ (spatial cutoff) and

$\hat{P}_{\leq \Lambda} = F P_{\leq \Lambda} F^{-1}$ (Fourier/momentum cutoff) to define compressed operators $R = \hat{P} P \hat{P}$ on Hilbert spaces $L^2(X_S)$ over local/semi-local fields¹⁷.

- *Reusability Potential: Moderate.* Provides a rigorous method for regularizing continuous spectrum on non-compact spaces into discrete or compressed representations¹⁷. However, it is fundamentally tied to multiplicative scaling flows rather than additive time evolution along deterministic trajectories¹⁷.

2. Oscillator Basis Projections on $\mathbb{Q}_p$ (Srednicki)²⁹:

- *Technique:* Projecting differential operators (such as x^p) onto finite-level subspaces spanned by p -adic harmonic oscillator wavefunctions $\psi_{v,N}(x)$ ²⁹.
- *Reusability Potential: High (Local Only).* Provides a template for constructing linear matrix representations of operators acting on $L^2(\mathbb{Q}_p)$, provided the system can be mapped to an ultrametric harmonic oscillator basis²⁹.

3. Limit-Periodic Jacobi Operators on p -Adic Hulls (Peherstorfer-Yuditskii-Volberg)⁴⁴:

- *Technique:* Translating deterministic flows on p -adic integer hulls $\mathbb{Z}_p$ into tridiagonal Jacobi operators via orthogonal polynomials over Cantor measures⁴².
- *Reusability Potential: High.* This machinery is directly capable of taking a deterministic, p -adically limit-periodic dynamical system and constructing a self-adjoint operator with an explicit singular continuous spectrum⁴⁴.

Final Explicit Verdict

SAME CATEGORY OF MATHEMATICAL OBJECT, NO WORKING BRIDGE YET CONSTRUCTED BY ANYONE.

The technical rationale for this verdict rests on three fundamental structural divergences:

1. **Symmetries Misalignment:** The mathematical machinery of Tate-Iwasawa theory and Connes' program relies intrinsically on **multiplicative/automorphic symmetries** (the scaling action of ideles $\mathbb{I}_{\mathbb{Q}}$ and Poisson summation over global field embeddings $\mathbb{Q} \subset \mathbb{A}_{\mathbb{Q}}$)¹. In contrast, a Koopman operator arising from deterministic physical dynamics relies on **translation flows along vector fields** or continuous endomorphisms on phase space¹¹. No existing theorem bridges the transformation of idele class scaling flows into continuous vector-field Koopman generators on fractal invariant sets¹¹.
2. **Divergent Spectral Targets:** The machinery developed around the Hilbert-Pólya conjecture is strictly optimized to produce or analyze **pure point spectra** (or trace-formula representations of point zeros)¹⁷. Conversely, deterministic dynamics on fractal/adelic substrates natively generate **singular continuous spectra**⁴³. Consequently,

the spectral regularization tools of Connes and Berry-Keating are designed to eliminate or compress the continuous spectrum, making them structurally incompatible with systems where singular continuous measures are the desired physical feature¹⁷.

3. **Absence of Domain-Preserving Koopman Constructions:** While ergodic theory on $\mathbb{P}$

-adic balls (Anashin et al.) and $\mathbb{P}$ -adic random matrix theory (Van Peski) exist, no established literature demonstrates how to extract a closed, self-adjoint, second-order

linear operator subspace (a quantum-amplitude layer) from a deterministic $\mathbb{P}$ -adic/adelic Koopman operator without breaking measure preservation³².

While candidate tools—specifically Peherstorfer-Yuditskii-Volberg Jacobi matrices⁴⁴ and

Srednicki's $\mathbb{P}$ -adic projection techniques²⁹—provide relevant mathematical building blocks, adapting them to build a self-adjoint Koopman spectral generator for Bohmian mechanics on an adelic invariant set requires constructing an entirely new mathematical bridge.

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